

# Linear regression: Abalone growth and pH example

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## Set up

```
library(tidyverse) # general use
library(janitor) # cleaning data frames
library(here) # file/folder organization
library(readxl) # reading .xlsx files
library(ggeffects) # generating model predictions
library(flextable) # creating tables
```

```
# abalone data from Hamilton et al. 2022
abalone <- read_xlsx(here("data", "Abalone IMTA_growth and pH.xlsx"))
```

## Abalone example

Data from Hamilton et al. 2022. “Integrated multi-trophic aquaculture mitigates the effects of ocean acidification: Seaweeds raise system pH and improve growth of juvenile abalone.”  
<https://doi.org/10.1016/j.aquaculture.2022.738571>

### a. Questions and hypotheses

Question: How does pH predict abalone growth (measured in change in shell surface area per day,  $\text{mm}^2 \text{d}^{-1}$ )?

$H_0$ : pH has no effect on abalone shell growth.

$H_A$ : pH does have an effect on abalone shell growth.

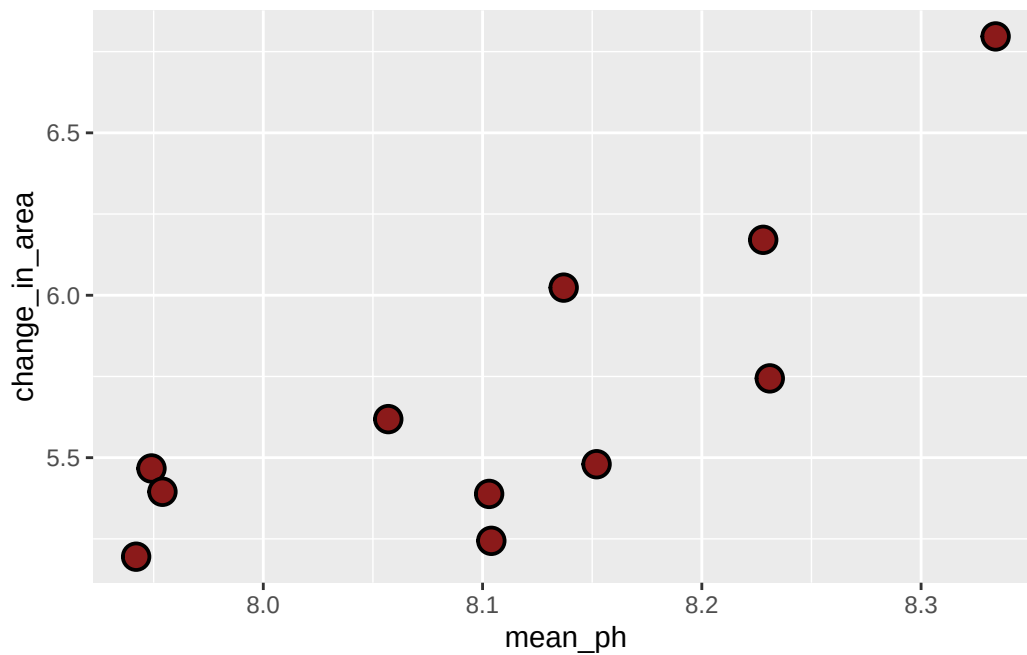
### b. Cleaning

```
# creating new object from original data frame
abalone_clean <- abalone |>
  # cleaning names to remove spaces
  clean_names() |>
  # selecting columns of interest
  select(mean_p_h, change_in_area_mm2_d_1_25) |>
  # renaming columns
  rename(mean_ph = mean_p_h,
         change_in_area = change_in_area_mm2_d_1_25)
```

Don't forget to look at your data! Use `View(abalone_clean)` in the *Console*.

### c. Exploratory data visualization

```
# base layer: ggplot
ggplot(data = abalone_clean, # adding data
       aes(x = mean_ph, # x-axis: pH, y-axis: change in shell surface area/day
           y = change_in_area)) +
# second layer: points to represent observations
geom_point(size = 4,
           stroke = 1,
           fill = "firebrick4",
           shape = 21)
```



Where does the missing value come from?

**The missing value comes from one pair of tanks in which the pH sensor malfunctioned.**

Don't forget to turn your warnings off either in the individual code chunk or in the YAML at the top of the file.

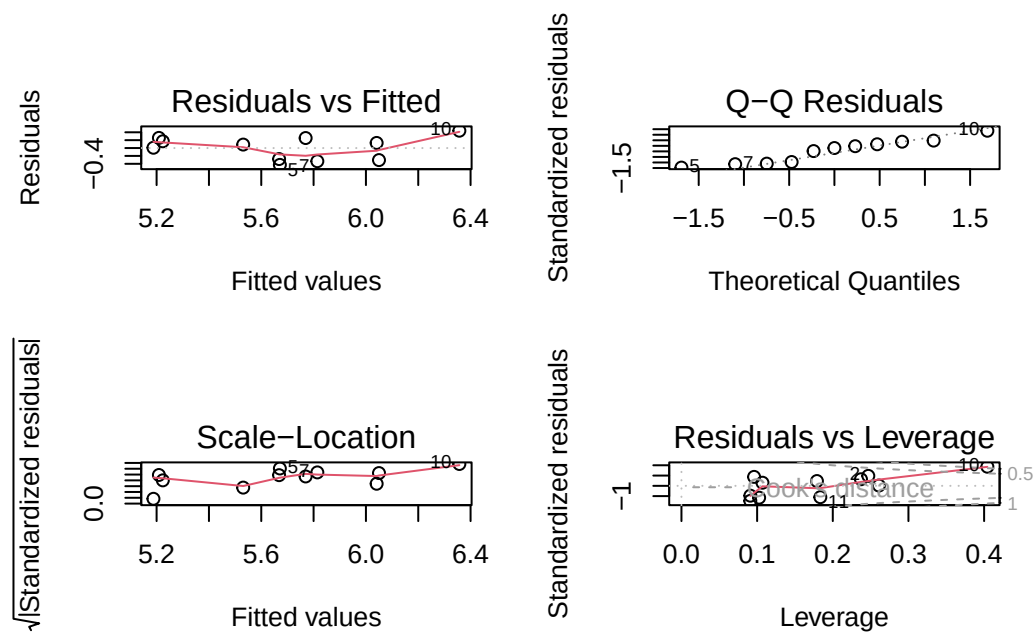
#### d. Abalone model

##### Model fitting

```
abalone_model <- lm(
  change_in_area ~ mean_ph, # formula: change in shell SA/day as a function of pH
  data = abalone_clean      # data frame: cleaned abalone data
)
```

## Diagnostics

```
par(mfrow = c(2, 2)) # displays plots in a 2x2 grid
plot(abalone_model)   # displays diagnostic plots for model
```



## Model summary

```
# more information about the model
summary(abalone_model)
```

Call:

```
lm(formula = change_in_area ~ mean_ph, data = abalone_clean)
```

Residuals:

|  | Min      | 1Q       | Median  | 3Q      | Max     |
|--|----------|----------|---------|---------|---------|
|  | -0.42714 | -0.29282 | 0.08769 | 0.21258 | 0.43965 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )   |
|-------------|----------|------------|---------|------------|
| (Intercept) | -18.5010 | 6.1601     | -3.003  | 0.01488 *  |
| mean_ph     | 2.9827   | 0.7596     | 3.926   | 0.00348 ** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3063 on 9 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.6314, Adjusted R-squared: 0.5905

F-statistic: 15.42 on 1 and 9 DF, p-value: 0.003477

What is the slope estimate (with standard error?)

$\beta_1$ :  $2.98 \pm 0.76$

What is the intercept estimate (with standard error?)

$\beta_0$ :  $-18.50 \pm 6.16$

## Generating predictions

```
# creating a new object called abalone_preds
abalone_preds <- ggpredict(
  abalone_model,      # model object
  terms = "mean_ph"   # predictor (in quotation marks)
)

# display the predictions
abalone_preds
```

# Predicted values of change\_in\_area

| mean_ph | Predicted | 95% CI     |
|---------|-----------|------------|
| 7.94    | 5.19      | 4.83, 5.54 |
| 7.95    | 5.21      | 4.86, 5.55 |
| 8.06    | 5.53      | 5.30, 5.76 |

|      |  |      |  |            |
|------|--|------|--|------------|
| 8.10 |  | 5.67 |  | 5.46, 5.88 |
| 8.10 |  | 5.67 |  | 5.46, 5.88 |
| 8.14 |  | 5.77 |  | 5.55, 5.98 |
| 8.15 |  | 5.81 |  | 5.59, 6.04 |
| 8.33 |  | 6.36 |  | 5.92, 6.80 |

Look at the column names using `colnames(abalone_preds)` in the Console.

What are the column names?

`x`, `predicted`, `std.error`, `conf.low`, `conf.high` (not going to worry about `group` right now)

Look at the actual object (by clicking on it in Environment).

Look at the class of the object using `class(abalone_preds)` in the Console.

What is this type of object?

It is a `ggeffects` output and a regular data frame.

View `abalone_preds` like a data frame.

### Why generate model predictions?

This dataset does not include any *actual* observations of abalone growth at  $\text{pH} = 8$ .

However, the model can tell us a *prediction* for what abalone growth could be if  $\text{pH} = 8$  based on the existing data.

Figure out what abalone growth the model would *predict* at  $\text{pH} = 8$ .

```
# finding the model prediction at a specific value
ggpredict(
  abalone_model,      # model object
  terms = "mean_ph[8]" # predictor (in quotation marks) and predictor value in brackets
)
```

# Predicted values of `change_in_area`

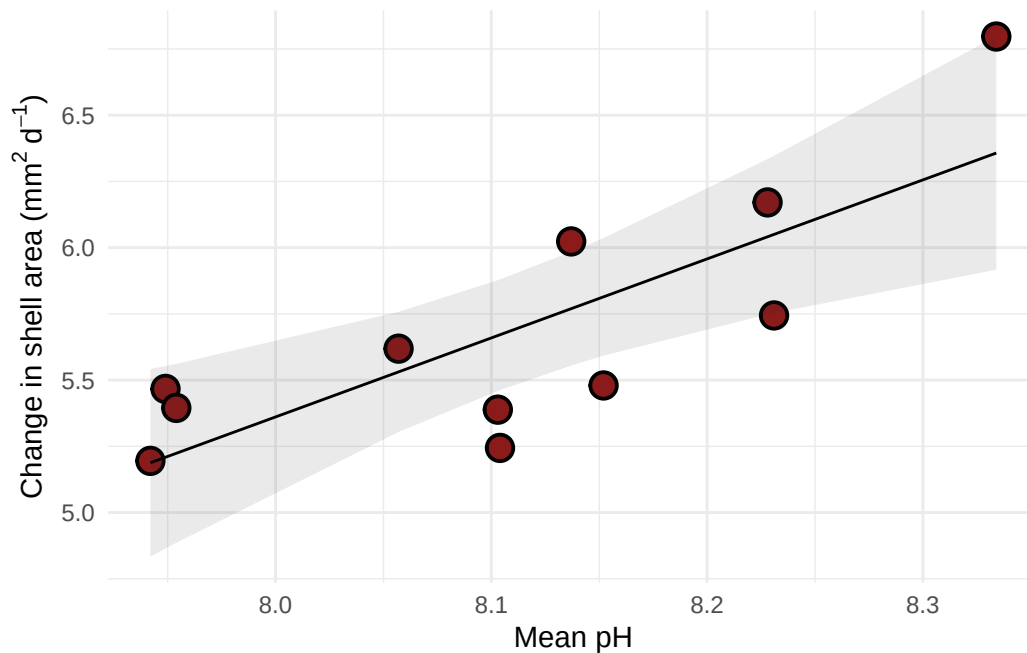
| mean_ph |  | Predicted |  | 95% CI     |
|---------|--|-----------|--|------------|
| 8       |  | 5.36      |  | 5.08, 5.64 |

How would you interpret this?

**At  $\text{pH} = 8$ , the model predicts change in abalone shell surface area to be 5.36 [95% CI: 5.08, 5.64]  $\text{mm}^2 \text{d}^{-1}$ .**

## Visualizing model predictions and data

```
# base layer: ggplot
# using clean data frame
ggplot(data = abalone_clean,
       aes(x = mean_ph,
           y = change_in_area)) +
# first layer: points representing abalones
geom_point(size = 4,
           stroke = 1,
           fill = "firebrick4",
           shape = 21) +
# second layer: ribbon representing confidence interval
# using predictions data frame
geom_ribbon(data = abalone_preds,
          aes(x = x,
              y = predicted,
              ymin = conf.low,
              ymax = conf.high),
          alpha = 0.1) +
# third layer: line representing model predictions
# using predictions data frame
geom_line(data = abalone_preds,
         aes(x = x,
             y = predicted)) +
# axis labels
labs(x = "Mean pH",
     y = expression("Change in shell area (*mm2~d-1")) +
# theme
theme_minimal()
```



**Creating a table with model coefficients, 95% confidence intervals, and more**

**Note:** this function is from `flextable`.

```
as_flextable(abalone_model)
```

|             | Estimate | Standard Error | t value | Pr(> t ) |    |
|-------------|----------|----------------|---------|----------|----|
| (Intercept) | -18.501  | 6.160          | -3.003  | 0.0149   | *  |
| mean_ph     | 2.983    | 0.760          | 3.926   | 0.0035   | ** |

*Signif. codes: 0 <= '\*\*\*' < 0.001 < '\*\*' < 0.01 < '\*' < 0.05*

Residual standard error: 0.3063 on 9 degrees of freedom

Multiple R-squared: 0.6314, Adjusted R-squared: 0.5905

F-statistic: 15.42 on 9 and 1 DF, p-value: 0.0035

**END OF ABALONE EXAMPLE**